

Basics

Classical Information — Introduction to Quantum Information

Suggested time: **60 minutes** | Reading: Nielsen & Chuang 11.1–11.2, Preskill Ch. 5.1

Notation reminder

- X is a random variable; x is a value it may take, from a set of d possibilities, with probability $p(x)$.
- $K_N[X]$ = *Kolmogorov complexity*: the number of bits needed to store N outcomes of X after optimal compression.
- $H[X] = -\sum_x p(x) \log_2 p(x)$ = *Shannon entropy*: bits per outcome, i.e. $\lim_{N \rightarrow \infty} K_N[X]/N$.
- $I(Y; Z) = H(Y) + H(Z) - H(Y, Z)$ = *mutual information*.
- Binary case: $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$.

Useful values: $\log_2 3 = 1.585$, $\log_2 6 = 2.585$, $\log_2 27 = 4.755$, $\log_2 128 = 7$.

Timing guide: Part A $\approx$ 15 min Part B $\approx$ 30 min Part C $\approx$ 15 min.

Part A — Information, alphabets and encoding

1. Counting characters and counting bits. A message is written in the Latin alphabet (upper and lower case, digits, punctuation — about 100 symbols in total) and stored on a computer using 7-bit ASCII.

- Explain in one sentence why the conversion factor is $\log(128)/\log(2) = 7$ bits per Latin character.
- A message contains 500 Latin characters. How many binary characters (bits) does it occupy?
- Now suppose we restrict ourselves to the 27-symbol alphabet $\{\mathbf{a-z, space}\}$. What is the smallest number of bits per character achievable with a *fixed-length* binary code? What does $\log_2 27$ give, and why are the two answers different?

2. A variable-length code: Morse. Written in binary (1 = signal on for one time unit, 0 = off), Morse code gives

$$\mathbf{A} = 10111, \quad \mathbf{B} = 111010101, \quad \mathbf{AB} = 1011100111010101.$$

- How many bits does **AB** use in Morse, and how many in 7-bit ASCII? Which wins here?
- Identify the 00 in the middle of **AB**. What is it for, and what would go wrong without it?
- In Morse, **E** (the most common English letter) is a single dot while **Q** is long. Explain qualitatively why a code that gives frequent symbols short strings beats a fixed-length code on real English text. (Keep this in mind for Q6 — it is the whole idea behind Shannon entropy.)

3. Why “number of characters” is a bad measure of information. Consider N pages, each containing nothing but 3000 zeros.

- How many characters does this object contain, written out in full?
- The whole thing is completely specified by the sentence “ N pages, each containing nothing but 3000 zeros”, which is about 35 Latin characters plus however many are needed to write the number N . Take $N = 10^6$ and estimate the total length of this description in Latin characters.
- Hence estimate the *information per page*, and show that it tends to 0 as $N \rightarrow \infty$. Contrast this with a well-written textbook of N pages.

Part B — Shannon entropy

4. Your first entropy calculation. A biased binary source emits 0 with probability $p(0) = 0.9$ and 1 with probability $p(1) = 0.1$, independently each time (i.i.d.).

- Compute $H[X]$ in bits.
- How many bits are needed, asymptotically, to store 1000 outcomes after optimal compression?
- What compression ratio does this represent, compared with just printing the string as it is?

5. The uniform distribution.

- Compute the Shannon entropy of a single throw of a fair six-sided die, in bits.
- Show that for a uniform distribution over d outcomes, $H = \log_2 d$ bits.
- One of the three conditions we demand of a measure of randomness is that it be *maximised* by the uniform distribution. Explain in words why that is the right thing to demand.

6. Entropy as average surprise. Define the surprise of an outcome as $S(x) = -\log_2 p(x)$.

- Check that $S(x) = 0$ when $p(x) = 1$ and $S(x) \rightarrow \infty$ as $p(x) \rightarrow 0$. Why are these the behaviours we want?
- For two independent events we require $S(x_1, x_2) = S(x_1) + S(x_2)$. Show that $S = -\log p$ satisfies this, and argue briefly why this requirement essentially forces the logarithm on us.
- Show that $\langle S(x) \rangle = \sum_x p(x) S(x) = H[X]$.
- Compute the surprise of rolling a 6 with (i) a fair die, (ii) a loaded die with $p(6) = 1/2$.

7. The binary entropy function. Let $p(0) = p$ and $p(1) = 1 - p$, so that $H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$.

- Evaluate $H(p)$ at $p = 0, 1/4, 1/2, 3/4, 1$ and sketch the curve on $0 \leq p \leq 1$.
- By differentiating, show that the maximum is at $p = 1/2$ and find its value.
- Explain what $H(0) = H(1) = 0$ means operationally, in terms of storing N outcomes.

8. Compression of a sparse source. A source emits 1 with small probability $p(1) = 0.01$ and 0 otherwise, i.i.d. A natural scheme is to record only the *lengths of the runs of consecutive zeros*, e.g. 00000 1 0000000 1 00 1 $\rightarrow 5, 7, 2$.

- What is the average length of a run of zeros?
- Roughly how many bits are needed to record one such run length?
- Roughly how many runs occur in N outcomes? Hence estimate K_N and the bits per outcome.
- Compare your estimate with the exact value $H(0.01)$. Is your estimate above or below, and why?

Part C — From independence to shared information

9. Additivity for independent variables. Let X be the composite variable made of two *independent* variables Y and Z , so that $p(y, z) = p(y)p(z)$. Show that

$$H[p(y, z)] = H[p(y)] + H[p(z)].$$

Hint: expand $\log_2(p(y)p(z))$, split the double sum, and use $\sum_y p(y) = 1$.

10. Shared information. Two coins, Y and Z , each with outcomes {heads, tails}. Consider three joint distributions:

- (i) independent fair coins: $p(y, z) = 1/4$ for all four pairs
- (ii) perfectly correlated coins: $p(HH) = p(TT) = 1/2$, others 0
- (iii) ★ partially correlated: $p(HH) = p(HT) = p(TH) = 1/3$, $p(TT) = 0$

For each case, find the marginals $p(y)$ and $p(z)$, then compute $H(Y)$, $H(Z)$, $H(Y, Z)$ and $I(Y; Z) = H(Y) + H(Z) - H(Y, Z)$. Part (iii) is optional if you are short of time.

Check that your answer to case (i) is consistent with what you proved in Q9. Then interpret case (ii) in one sentence: both coins store a bit — how many bits are there in total, and why?

Every quantity on this sheet has a quantum counterpart: $H[X]$ becomes the von Neumann entropy $S(\rho)$, and $I(Y; Z)$ becomes the quantum mutual information — which, unlike here, can exceed what any classical pair of variables is allowed to share. Keep this sheet.